Font-End Technologies (HTML)

BIS605

Software Development Technologies for Digital Business

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References

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HTML

- Standard markup language for documents designed to be displayed in a web browser.

- Web browser uses HTML to interpret and compose text, images, and other material into visual or audible web pages.

- HTML provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes and other items.

- HTML elements are the building blocks of HTML pages which delineated by *tags*, written using angle brackets <>.

HTML

- HTML can embed programs written in a scripting language such as JavaScript, which affects the behavior and content of web pages.

- Inclusion of CSS defines the look and layout of content.

HTML History

- HTML 1.0 – Limited features for designing your web pages
- HTML 2.0 – HTML 1.0 + new features for web page design.
- HTML 3.0 – included many new useful enhancements to HTML. However, most browser implemented only few elements from this draft.
- HTML 3.2 (WILBUR) – Most, if not all, popular browsers in use today fully support HTML 3.2
- HTML 4.0 (COUGAR) – introduces new functionality, unfortunately Netscape does not recognize many tags and attributes.
- XHTML - standard to provide rich high quality web pages through these varied devices

HTML 5

- HTML 5 is the new web standard. It follows HTML 4 and XHTML
- The basic aim of HTML5 is to provide two things
  - to improve the language
  - to support the latest multimedia.
- Among them were to reduce the need for external plug-ins (such as Flash plug-ins), better handling of errors, and more markup elements (tags) to replace scripting.
- HTML5 should also be device independent while also keeping it easily readable by us humans.

HTML 5

Type of content	HTML 1.2	HTML 4.01	HTML5	Purpose
Heading	Yes	Yes	Yes	Organize page content by adding headings and subheadings to the top of each section of the page
Paragraph	Yes	Yes	Yes	Identify paragraphs of text
Address	Yes	Yes	Yes	Identify a block of text that contains contact information
Anchor	Yes	Yes	Yes	Link to other web content
List	Yes	Yes	Yes	Organize items into a list
Image	Yes	Yes	Yes	Embed a photograph or drawing into a web page
Table	No	Yes	Yes	Organize data into rows and columns
Style	No	Yes	Yes	Add CSS to control how objects on a web page are presented
Script	No	Yes	Yes	Add Javascript to make pages respond to user behaviors (more interactive)
Audio	No	No	Yes	Add audio to a web page with a single tag
Video	No	No	Yes	Add video to a web page with a single tag
Canvas	No	No	Yes	Add an invisible drawing pad to a web page, on which you can add drawings (animations, games, and other interactive features) using Javascript

HTML Elements

- An HTML element is defined by a start tag, some content, and an end tag.

<tagname> Content…..</tagname>

- HTML elements can be nested

```
<!DOCTYPE html>
<html>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

HTML Attributes

- All HTML elements can have attributes
- Attributes provide additional information about elements
- Attributes are always specified in the start tag
- Attributes usually come in name/value pairs like: name="value"

```html
<a href="https://www.w3schools.com">Visit W3Schools</a>
```

Visit W3Schools

HTML Global Structure

index.html

index.html > html > body > p

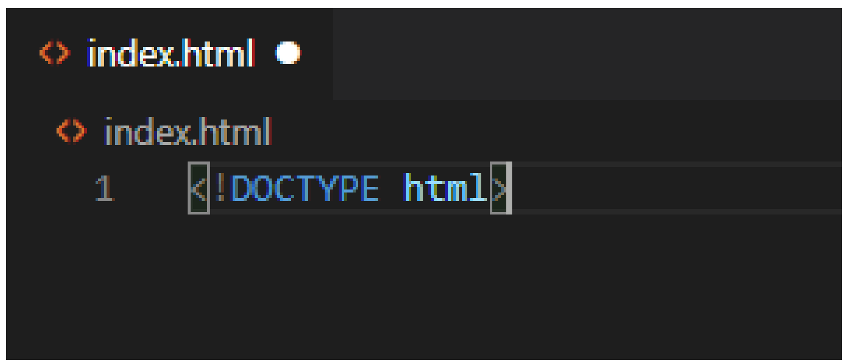
```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Title of the document</title>
    <meta name="description" content="Index Web Page">
    <meta name="keywords" content="BIS605, Web-based Development, HTML, CSS, JavaScript">
    <meta name="author" content="Worarat Krathu">
</head>
<body>

    <h1>This is a heading</h1>
    <p>This is a paragraph.</p>

</body>
</html>
```

Create a HTML Web page

- Steps:
  - Create a folder named "MyWebpages"
  - Open IDE or any text editor (notepad, notepad++). In this lab, we will use Visual Studio.
  - Drag the folder to the Explorer of the Visual Studio.
  - Right Click -> New File -> name the file as "index.html".

Doctype Tag

- All HTML must start with Doctype information
- It is not HTML tag, instead it is the information for browser regarding type of document.



HTML Tag

- <html> tag represents the root of HTML document.
- It is the container for all other HTML elements.
- The lang attribute specifies language of the Web page for search engines and browsers.

```
index.html ×
index.html > ...
1   <!DOCTYPE html>
2 > <html lang="en">···
15  </html>
```

Head Tag

- The <head> tag represents head section.
- It usually contains title and metadata of a Web page.
- The following elements can go inside the <head> element.
  - <title>, <style>, <link>, <meta>, <script>, <base>, <noscript>

index.html ×

index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Title of the document</title>
    <meta name="description" content="Index Web Page">
    <meta name="keywords" content="BIS605, Web-based Development, HTML, CSS, JavaScript">
    <meta name="author" content="Worarat Krathu">
</head>
```

Meta Tag

- The <meta> tag provides information (metadata) of a Web page.
- It will not be displayed on the Web page.
- It is used by browsers, search engines, and other web services.

index.html ×

index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Title of the document</title>
    <meta name="description" content="Index Web Page">
    <meta name="keywords" content="BIS605, Web-based Development, HTML, CSS, JavaScript">
    <meta name="author" content="Worarat Krathu">
</head>
```

Body Tag

- The <body> tag defines the body of Web page.
- It can contains other elements representing all contents of HTML document such as headings, paragraphs, images, hyperlinks, tables, lists, etc.

index.html

index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Title of the document</title>
    <meta name="description" content="Index Web Page">
    <meta name="keywords" content="BIS605, Web-based Development, HTML, CSS, JavaScript">
    <meta name="author" content="Worarat Krathu">
</head>
<body>

    <h1>This is a heading</h1>
    <p>This is a paragraph.</p>

</body>
</html>
```

Comment Tag

- An exclamation point (!) is used in the start tag but not the end tag.

```
<body>
    <!--This is comments-->
    <h1>This is a heading</h1>
    <p>This is a paragraph.</p>

</body>
</html>
```

Exercise 7.1

- Build HTML Structure in the “index.html” by inserting the following tag:
  - <!DOCTYPE html>
  - <html>
  - <head>
  - <title>
  - <meta>
  - <body>
- Then put appropriate attribute for <html>, <title>, and <meta> tag.

Exercise 7.2

- Explore the following tags, use them in your “index.html” and describe function of each tag.
  - <h1>, <h2>, <h3>, <h4>, <h5>, <h6>
  - <p> and
  - <a>
  - <img>, <figure>, <figcaption>
  - <audio>, <source>
  - <video>, <source>
  - <iframe>

Exercise 7.3

- Explore formatting elements to display special type of text and describe function of each tag.
  - <b>
  - <strong>
  - <i>
  - <em>
  - <mark>
  - <small>
  - <del>
  - <ins>
  - <sub>
  - <sup>

Exercise 7.4

- Explore the following tags, use them in your “index.html” and describe function of each tag.
  - <table>, <tr>, <td>
  - <thead>, <tfoot>
  - ,
  - ,
  - <dl>,<dt>,<dd>

Form Elements

- HTML Forms are used to collect input from user. Those inputs are commonly sent to server and processed at the server side.

- The <form> tag is used to open and close the scope of form.

```
<form>

</form>
```

Label

- The <label> element defines a label for several form elements. It appears as label.

Label

```
<label>Label</label>
```

- Label can be pair with other input elements by using attribute for.

Login Button Log in

```
<label for="loginbutton">Login Button</label>
<input type="button" id="loginbutton"
value="Log in">
```

Form Elements

- The <input> tag represents input for a form.
- There are several types of input which defined by the type attribute. Default type is “text”.

  - <input type="button">
  - <input type="checkbox">
  - <input type="color">
  - <input type="date">
  - <input type="datetime-local">
  - <input type="email">
  - <input type="file">
  - <input type="hidden">
  - <input type="image">
  - <input type="month">
  - <input type="number">
  - <input type="password">
  - <input type="radio">
  - <input type="range">
  - <input type="reset">
  - <input type="search">
  - <input type="submit">
  - <input type="tel">
  - <input type="text">
  - <input type="time">
  - <input type="url">
  - <input type="week">

Button Element

- There are two ways to create a button
  - Use the <input> tag, <input type="button">
  - Use the <button> tag which allows to have content inside such as text or images, etc.

```
<input type="button" value="Log in">
```


```
<button type="buttion">
    <img src="loginbutton.png" height="30"
    width="80">
</button>
```

Checkbox

- Input type “checkbox” defines a checkbox. It can be defined using attribute “type” in the <input> tag.

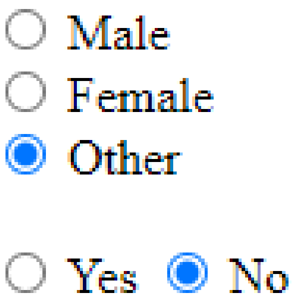
☐ Orange ☐ Strawberry ☐ Cherry

```
<input type="checkbox" id="orange" value="Orange">
<label for="orange">Orange</label>
<input type="checkbox" id="strawberry" value="Strawberry">
<label for="strawberry">Strawberry</label>
<input type="checkbox" id="cherry" value="Cherry" >
<label for="cherry">Cherry</label>
```

Radio

- The input type “radio” defines radio buttons where only one choice can be selected. A group of radio buttons associated by “name” attribute.

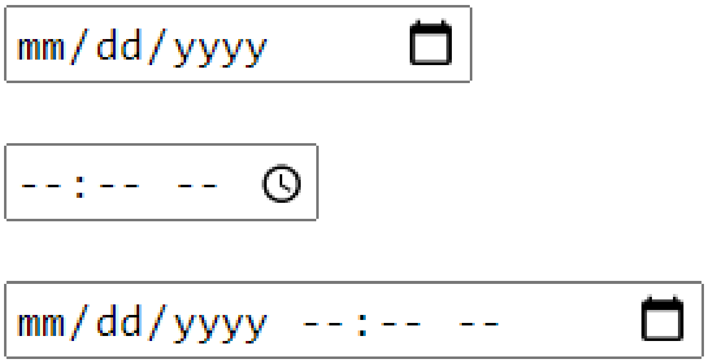
```html
<input type="radio" id="male" name="gender" value="male">
<label for="male">Male</label><br>
<input type="radio" id="female" name="gender" value="female">
<label for="female">Female</label><br>
<input type="radio" id="other" name="gender" value="other">
<label for="other">Other</label>
<br>
<br>
<input type="radio" id="yes" name="hello" value="yes">
<label for="yes">Yes</label>
<input type="radio" id="no" name="hello" value="no">
<label for="no">No</label>
```



Date and Time

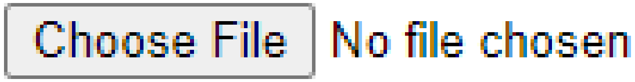
- The input type “date”, “time”, and “datetime-local” are used for inputs related to date and time. There will be date/time picker for the input.

```html
<input type="date">
<br>
<br>
<input type="time">
<br>
<br>
<input type="datetime-local">
```



File

- The input type “file” will define the field for file input. It will provide browse button to browse file from file location.




```
<input type="file">
```

Text and Password

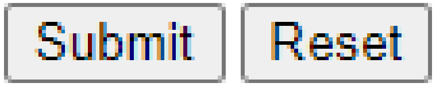
- The input type “text” and “password” define input of type text but for password the text will be hidden.

```html
<input type="text">
```

```html
<input type="password">
```

Reset and Submit

- The input of type “reset” and “submit” are visualized as a button.
- The reset type is used to reset a corresponding form.
- The submit type is used to submit data from all inputs in a corresponding form.




```html
<input type="submit">
<input type="reset">
```

Hidden

- The input “hidden” type defines hidden input field. It will not appear in a web page.
- The purpose of hidden input field is to allow a web page include data that cannot be seen by users when a form is submitted.

```html
<input type="hidden" id="testinput" name="testinput" value="1234">
```

Attribute

- Attributes in HTML tag provide additional information about elements.
- They are always used in the start tag and usually come as a pair of name and value.
- W3C recommends using quotes for values of attributes. Both single quote and double quote can be used. However, if a value contains double quotes, it is necessary to use single quote.

```
<a href='https://sit.kmutt.ac.th'>SIT Link to Website</a>
```

```
<img src="sample_image.jpg">
```

Global Attributes

- The global attributes can be used with all HTML elements.
- Some common global attributes:
  - class – specifies classnames for an element (normally it is a class in style sheet)
  - id – specifies unique id for an element
  - lang – specifies the language of the element's content
  - style – specifies an inline CSS style for an element
  - title – specifies extra information about an element (commonly appeared as tooltip)

Attributes

- List of attributes commonly used.
  - autoplay
  - control
  - loop
  - muted
  - src
  - colspan / rowspan
  - headers
  - height
  - width
  - href
  - name
  - list
  - disabled
  - checked
  - for
  - form
  - max / maxlength / min
  - method
  - multiple
  - required
  - Rows /cols
  - selected
  - size
  - type
  - value
  - wrap

Exercise 7.5

- Explore attributes from the previous slide. Try it in your html document and discuss what they specify and which tag they belong to.
- You can use comments to provide detail information of each attribute.

- For example:
  - href is an attribute specify URL of the page that the link goes to. It is used within the tag <a>
  - SIT Link to Website

Div Tag

- It is known as Division tag.
- It is used to make divisions of content in the web page like (text, images, header, footer, navigation bar, etc.).
  - <div> tag is Block level tag
  - It is a generic container tag
  - It is used to the group of various tags of HTML so that sections can be created and style can be applied to them.


```
<body>
  <div > div tag   </div>
  <div > div tag   </div>
  <div > div tag   </div>
  <div > div tag   </div>

  </body>
```

Layout Element

- HTML provides several semantic elements (elements clearly define their own content) to separate different parts of web page.

- Some semantic elements:
  - <header> - Defines a header for a document or a section
  - <nav> - Defines a set of navigation links
  - <section> - Defines a section in a document
  - <article> - Defines an independent, self-contained content
  - <aside> - Defines content aside from the content (like a sidebar)
  - <footer> - Defines a footer for a document or a section
  - <details> - Defines additional details that the user can open and close on demand
  - <summary> - Defines a heading for the <details> element

Header Tag

- The <header> tag in HTML is used to define the header for a document or a section.
  - contains information related to the title and heading of the related content.
  - commonly used for containing section's heading (h1-h6 elements) or wrap section's table of content, a search form, or any relevant logos.

This is the heading.

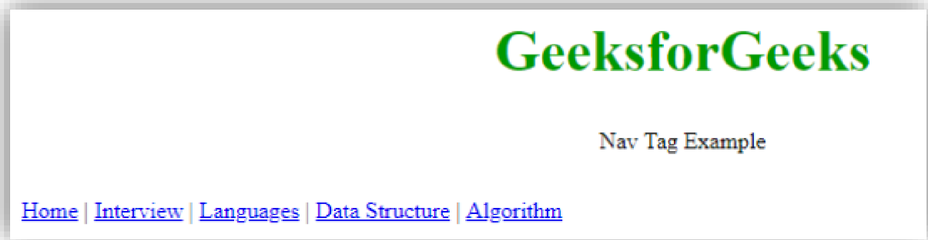
This is the sub-heading.

This is the metadata.

```html
<!DOCTYPE html>
<html>
    <head>
        <title>Header Tag</title>
    </head>
    <body>
        <article>
            <header>
                <h1>This is the heading.</h1>
                <h4>This is the sub-heading.</h4>
                <p>This is the metadata.</p>
            </header>
        </article>
    </body>
</html>
```

Nav Tag

- The <nav> tag defines a set of navigation links.
- It is intended only for major block of navigation links.




```html
<!DOCTYPE html>
<html>
    <head>
        <title>nav tag</title>
        <style>
            .gfg {
                font-size:40px;
                color:#090;
                font-weight:bold;
                text-align:center;
            }
            .nav_tag {
                text-align:center;
                margin:30px 0;
            }
        </style>
    </head>
    <body>
        <div class = "gfg">GeeksforGeeks</div>
        <div class = "nav_tag">Nav Tag Example</div>
        <nav>
            <a href = "#">Home</a> |
            <a href = "#">Interview</a> |
            <a href = "#">Languages</a> |
            <a href = "#">Data Structure</a> |
            <a href = "#">Algorithm</a>
        </nav>
    </body>
</html>
```

Section Tag

- It defines the section of documents such as chapters, headers, footers or any other sections.
- It is used when requirements of two headers or footers or any other section of documents needed.

Geeksforgeeek: Section 1

Content of section 1

GeeksforGeeks: Section 2

Content of section 2

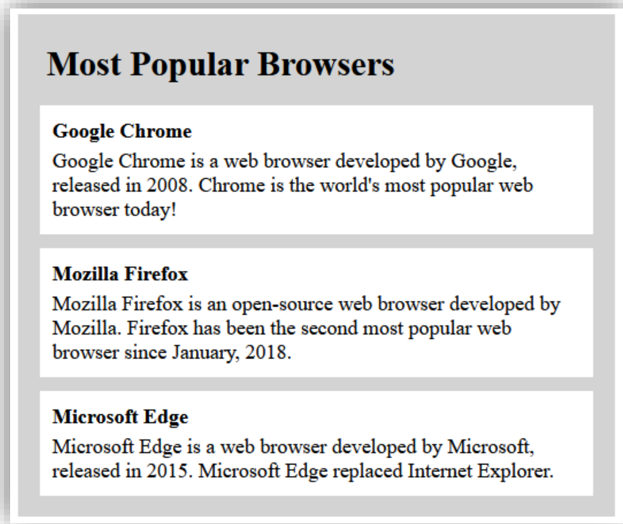
GeeksforGeeks: Section 3

Content of section 3

```html
<!DOCTYPE html>
<html>
    <head>
        <title>Section tag</title>
    </head>
    <body>
        <section>
            <h1>Geeksforgeeek: Section 1</h1>
            <p>Content of section 1</p>
        </section>
        <section>
            <h1>GeeksforGeeks: Section 2</h1>
            <p>Content of section 2</p>
        </section>
        <section>
            <h1>GeeksforGeeks: Section 3</h1>
            <p>Content of section 3</p>
        </section>
    </body>
</html>
```

Article Tag

- It represents a component of a page that consists of self-contained composition in a document, page or a site.
    - Forum post
    - Blog post
    - News story




```html
<article class="all-browsers">
  <h1>Most Popular Browsers</h1>
  <article class="browser">
    <h2>Google Chrome</h2>
    <p>Google Chrome is a web browser developed by Google,
released in 2008. Chrome is the world's most popular web
browser today!</p>
  </article>
  <article class="browser">
    <h2>Mozilla Firefox</h2>
    <p>Mozilla Firefox is an open-source web browser developed
by Mozilla. Firefox has been the second most popular web
browser since January, 2018.</p>
  </article>
  <article class="browser">
    <h2>Microsoft Edge</h2>
    <p>Microsoft Edge is a web browser developed by Microsoft,
released in 2015. Microsoft Edge replaced Internet
Explorer.</p>
  </article>
</article>
```

Aside Tag

- It basically identifies the content that is related to the primary content of the web page **but does not constitute the main intent** of the primary page.

```html
<body>
    <div class = "gfg">GeeksforGeeks</div>
    <article>
        <h1>Heading . . .</h1>
        <p>Aside tag is use to display important information
         about the primary page.</p>
    </article>
    <aside>
        <h1>Aside tag example</h1>
        <p>Aside tag content. . .</p>
    </aside>
</body>
```

Footer Tag

- It is used to define a footer of HTML document.
- It normally contains the footer information (author information, copyright information, carriers, etc).

```
        <footer>
            <nav>
                <p>
                    <a href=
"https://www.geeksforgeeks.org/about/">About Us</a>|
                    <a href=
"https://www.geeksforgeeks.org/privacy-policy/">Privacy Policy</a>|
                    <a href=
"https://www.geeksforgeeks.org/careers/">Careers</a>
                </p>
            </nav>
            <p>@geeksforgeeks, Some rights reserved</p>
        </footer>
```

About Us | Privacy Policy | Careers

@geeksforgeeks, Some rights reserved

Details and Summary Tag

- The <details> tag is used for the content/information which is initially hidden but could be displayed if the user wishes to see it.
- The <summary> tag is used to define a summary for the <details> element.


```html
<details>
    <summary>GeeksforGeeks</summary>
    <p>A computer science portal for geeks</p>
    <div>It is a computer science portal where you
    can learn programming.</div>
</details>
```

Layouts Techniques

- Using Tables,
- Using <div> and <span>
- CSS
  - CSS framework
  - CSS float property
  - CSS flexbox
  - CSS grid

Table Layout

```
<!DOCTYPE html>
<html>

   <head>
      <title>HTML Layout using Tables</title>
   </head>

   <body>
      <table width = "100%" border = "0">

         <tr>
            <td colspan = "2" bgcolor = "#b5dcb3">
               <h1>This is Web Page Main title</h1>
            </td>
         </tr>
         <tr valign = "top">
            <td bgcolor = "#aaa" width = "50">
               <b>Main Menu</b><br />
               HTML<br />
               PHP<br />
               PERL...
            </td>

            <td bgcolor = "#eee" width = "100" height = "200">
               Technical and Managerial Tutorials
            </td>
         </tr>
         <tr>
            <td colspan = "2" bgcolor = "#b5dcb3">
               <center>
                  Copyright © 2007 Tutorialspoint.com
               </center>
            </td>
         </tr>

      </table>
   </body>

</html>
```

Table Layout

```
<body>
   <table width = "100%" border = "0">

      <tr valign = "top">
         <td bgcolor = "#aaa" width = "20%">
            <b>Main Menu</b><br />
            HTML<br />
            PHP<br />
            PERL...
         </td>

         <td bgcolor = "#b5dcb3" height = "200" width = "60%">
            Technical and Managerial Tutorials
         </td>

         <td bgcolor = "#aaa" width = "20%">
            <b>Right Menu</b><br />
            HTML<br />
            PHP<br />
            PERL...
         </td>
      </tr>

   <table>
</body>
```

Main Menu HTML PHP PERL...	Technical and Managerial Tutorials	**Right Menu** HTML PHP PERL...

DIV and SPAN Layout

```html
<body>
  <div style = "width:100%">

    <div style = "background-color:#b5dcb3; width:100%">
      <h1>This is Web Page Main title</h1>
    </div>

    <div style = "background-color:#aaa; height:200px; width:20%; float:left;">
      <div><b>Main Menu</b></div>
      HTML<br />
      PHP<br />
      PERL...
    </div>

    <div style = "background-color:#eee; height:200px; width:60%; float:left;" >
      <p>Technical and Managerial Tutorials</p>
    </div>

    <div style = "background-color:#aaa; height:200px; width:20%; float:right;">
      <div><b>Right Menu</b></div>
      HTML<br />
      PHP<br />
      PERL...
    </div>

    <div style = "background-color:#b5dcb3; clear:both">
      <center>
        Copyright © 2007 Tutorialspoint.com
      </center>
    </div>

  </div>
</body>
```

CSS Framework

- WE can use existing CSS Frameworks such as W3.CSS or Bootstrap (very popular).
- Bootstrap is a free front-end framework for faster and easier web development
- Bootstrap includes HTML and CSS based design templates for typography, forms, buttons, tables, navigation, modals, image and many others which easy to create responsive designs.

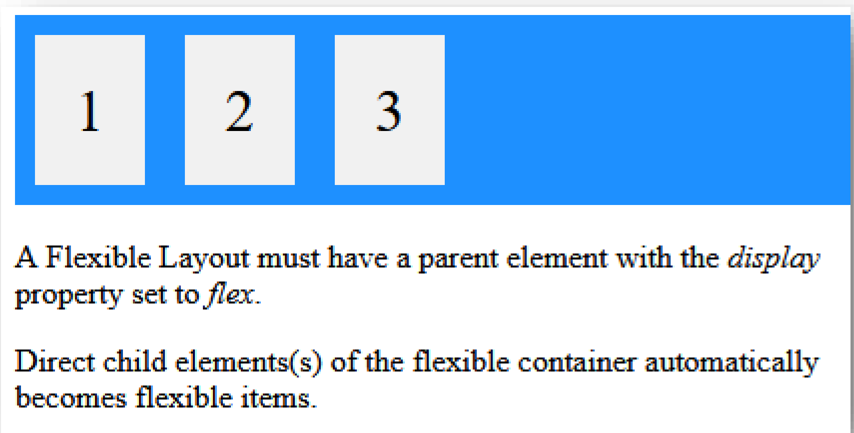
CSS Float Property

- Float property defines how element should float
  - none: This is default. The element does not float.
  - left: The element floats to the left of its container.
  - right: The element floats to the right of its container.
  - initial: set the property to its default.
  - inherit: inherits this property from its parent element.

```
<div style = "background-color:#aaa; height:200px; width:20%; float:left;">
```

CSS Flexbox

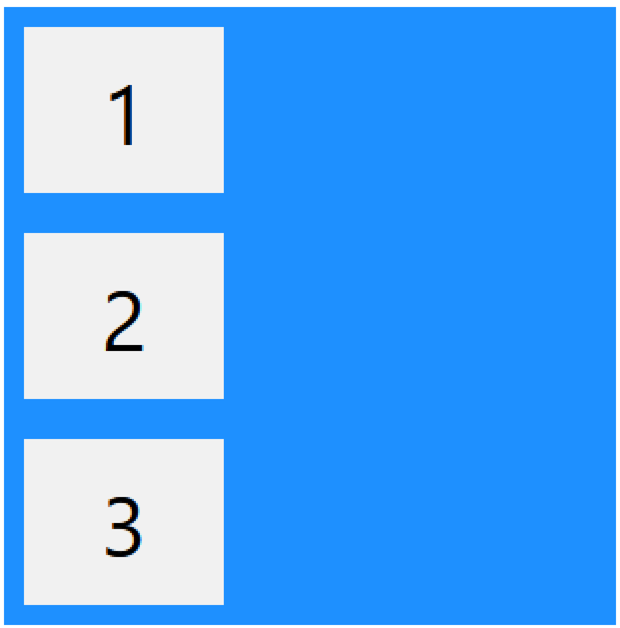
- The Flexible Box Layout Module, makes it easier to design flexible responsive layout structure without using float or positioning.




```html
<html>
<head>
<style>
.flex-container {
  display: flex;
  background-color: DodgerBlue;
}

.flex-container > div {
  background-color: #f1f1f1;
  margin: 10px;
  padding: 20px;
  font-size: 30px;
}
</style>
</head>
<body>

<div class="flex-container">
  <div>1</div>
  <div>2</div>
  <div>3</div>
</div>

<p>A Flexible Layout must have a parent element with the
<em>display</em> property set to <em>flex</em>.</p>

<p>Direct child elements(s) of the flexible container
automatically becomes flexible items.</p>

</body>
</html>
```

CSS Flexbox

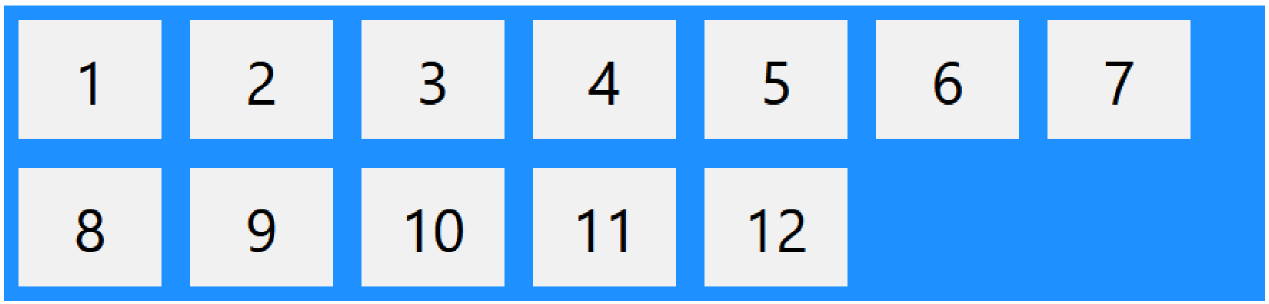
- There are several flex container properties.
  - flex-direction: stack flex items.




```css
.flex-container {
  display: flex;
  flex-direction: column;
}
```

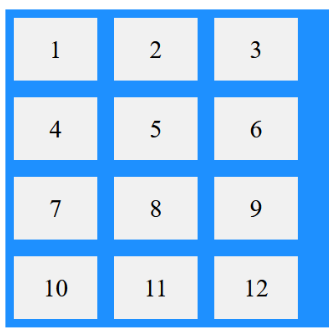
CSS Flexbox

- There are several flex container properties.
  - flex-wrap: wrap flex items if necessary.




```css
.flex-container {
  display: flex;
  flex-wrap: wrap;
}
```

  - flex-flow: shorthand property for flex-direction and flex-wrap



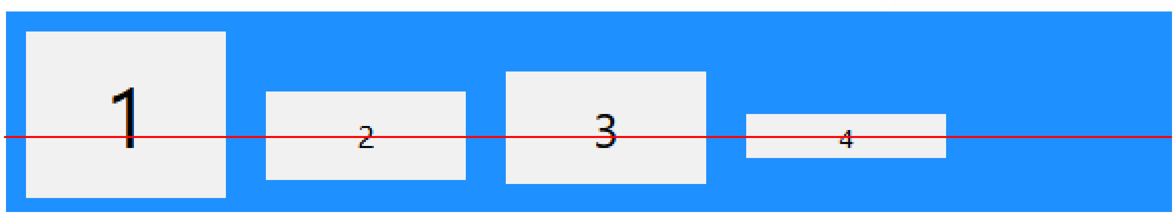

```css
.flex-container {
  display: flex;
  flex-flow: row wrap;
}
```

CSS Flexbox

- There are several flex container properties.
  - justify-content and align-items: align flex items.


```css
.flex-container {
  display: flex;
  justify-content: center;
}
```

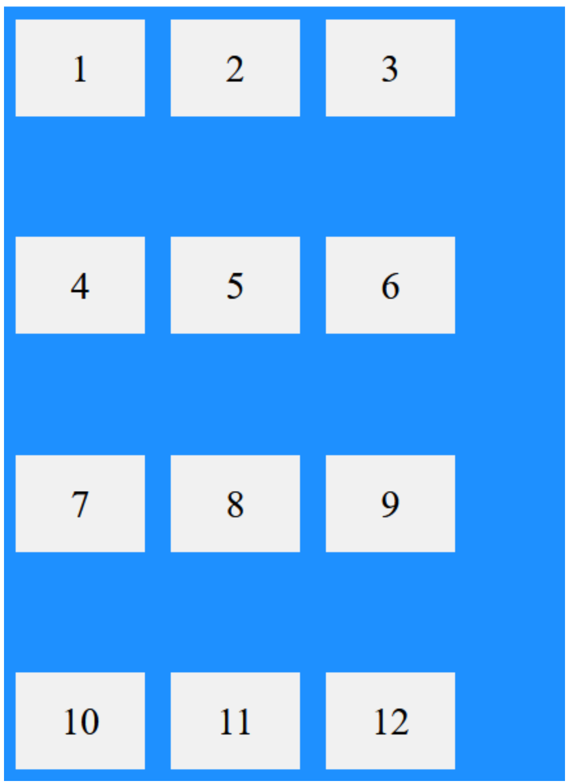
```css
.flex-container {
  display: flex;
  height: 200px;
  align-items: center;
}
```




```css
.flex-container {
  display: flex;
  height: 200px;
  align-items: baseline;
}
```

CSS Flexbox

- There are several flex container properties.
  - align-content: align flex lines.




```css
.flex-container {
  display: flex;
  height: 600px;
  flex-wrap: wrap;
  align-content: space-between;
  background-color: DodgerBlue;
}
```

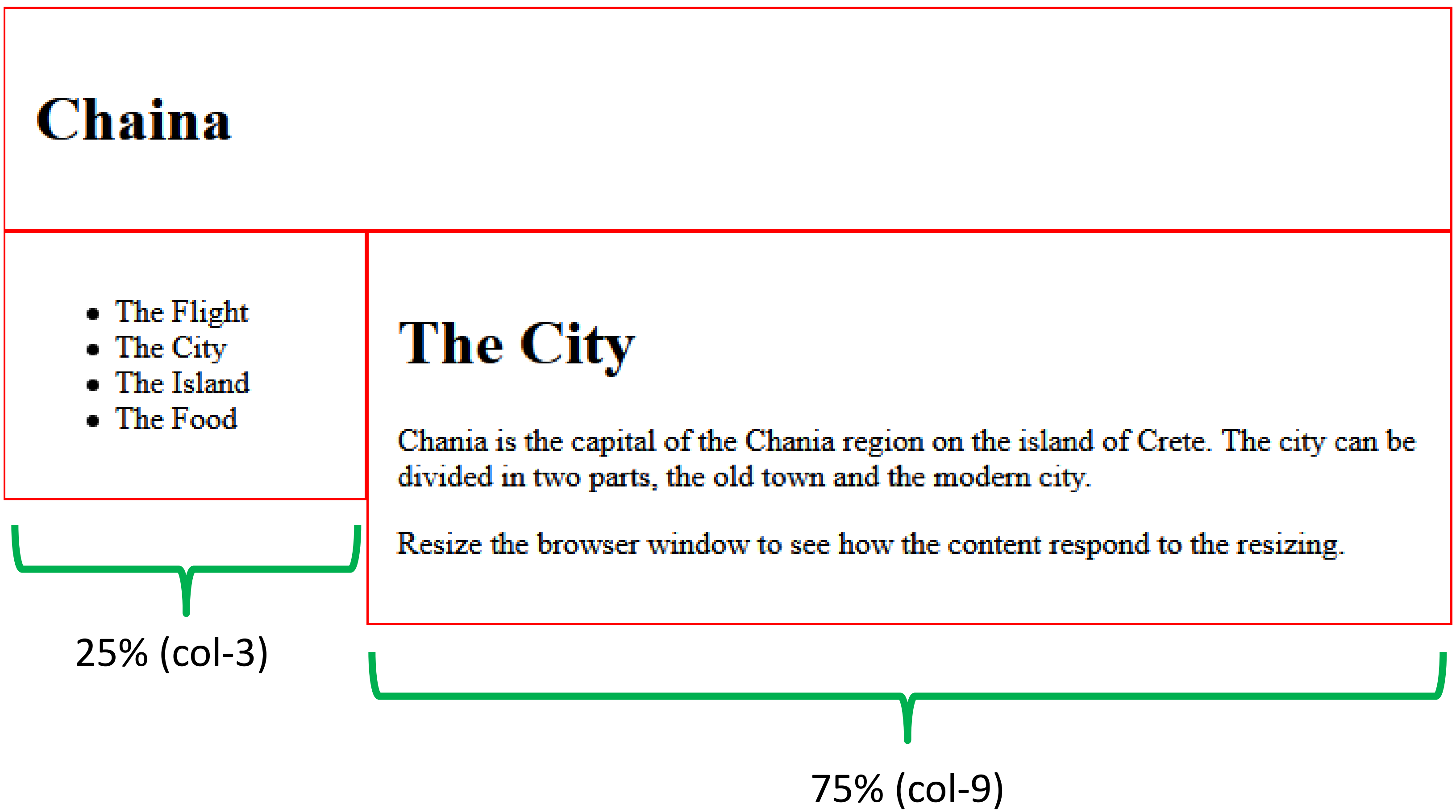
CSS Grid

- A responsive grid-view often has 12 columns, and has a total width of 100%, and will shrink and expand as you resize the browser window.

```html
<div class="col-3">
  <ul>
    <li>The Flight</li>
    <li>The City</li>
    <li>The Island</li>
    <li>The Food</li>
  </ul>
</div>

<div class="col-9">
  <h1>The City</h1>
  <p>Chania is the capital of the Chania region on the island of Crete. The
city can be divided in two parts, the old town and the modern city.</p>
  <p>Resize the browser window to see how the content respond to the
resizing.</p>
</div>
```

```css
.col-1 {width: 8.33%;}
.col-2 {width: 16.66%;}
.col-3 {width: 25%;}
.col-4 {width: 33.33%;}
.col-5 {width: 41.66%;}
.col-6 {width: 50%;}
.col-7 {width: 58.33%;}
.col-8 {width: 66.66%;}
.col-9 {width: 75%;}
.col-10 {width: 83.33%;}
.col-11 {width: 91.66%;}
.col-12 {width: 100%;}
```

CSS Grid


Chaina
The Flight
The City
The Island
The Food
The City
Chania is the capital of the Chania region on the island of Crete. The city can be divided in two parts, the old town and the modern city.
Resize the browser window to see how the content respond to the resizing.
25% (col-3)
75% (col-9)

Background Color / Width / Height

- Attributes: bgcolor, width, height are one of common attributes for setting layout.
- The bgcolor attribute is used for fill background color of the element. It uses html color codes as its value to display specific color.
- The height and the width attribute are used for setting the height and width of the elements. The value can be pixel or percentage of the device screen.

```
<td bgcolor = "#b5dcb3" height = "200" width = "60%">
```